Product Requirement Document (PRD): Real Estate CRM & Lead Management System

This document outlines the end-to-end product requirements, architecture specifications, schema design, and prompt operationalization for a modern Real Estate Lead Management System built for single-page artifact execution, Next.js deployment, and Supabase integration.

1. Project Overview & AI Coding Prompt Context

Executive Summary

The system provides a two-tier operational interface designed to bridge field data capture by Real Estate Consultants with executive oversight by System Admins. The objective is to eliminate lead drop-offs, standardize intake data across varied property types, and allow dynamic attribute tagging without requiring database schema migrations.

Recommended Technology Stack

Layer	Technology	Justification
Framework	Next.js 14+ (App Router) / React 18	Server-side rendering, API routes, optimized state handoff.
Styling	Tailwind CSS + Shadcn UI	Utility-first component framework with accessible primitive elements.
Icons	Lucide React	Lightweight, consistent SVG icon set.
State & Forms	React Hook Form + Zod	Schema-based client validation and fast rendering performance.
Database & Auth	Supabase (PostgreSQL)	Native Row-Level Security (RLS), real-time subscriptions, JSONB support.
Data Export	PapaParse / XLSX	Client-side data parsing and CSV transformation.

AI Code Generation Directives

When generating components via Claude Artifacts or LLM agents:

- Maintain all UI inside modular single-file React architectures where appropriate for Artifact rendering, utilizing standard Tailwind classes.
- Implement optimistic UI updates for lead entry and dynamic inline edits.
- Ensure dynamic schema inputs (JSONB key-value fields) validate properly prior to submission.
- Structure data calls using clean repository patterns or decoupled Supabase client functions.

2. User Roles & Authentication

Role-Based Access Control (RBAC) Matrix

Feature / Access Layer	Staff / Consultant	System Administrator
Authentication	Email / Password or Magic Link	Email / Password + Multi-Factor Authentication
Lead Capture Interface	Full Access (Create, Dynamic Tagging)	Access (For testing/manual entry)
Lead Scope	Restricted to self-created entries	Global (View, Edit, Delete all leads)
Field Customization	Append ad-hoc dynamic fields	Add/Modify global template fields
Inline Grid Editing	Disabled	Enabled (Real-time updates to DB)
Audit Logs & Export	No Access	Full Access (View audit trail, CSV export)

Functional Security Policies

- **Consultant Isolation**: Staff can query only rows matching `created_by == auth.uid()`.
- **Admin Privilege**: Elevated privileges grant access to global audit trails, bulk status updates, and soft-delete restorations.

3. Interface 1: Staff / Consultant Interface

The intake form prioritizes speed, mobile responsiveness, and input validation to allow field agents to capture multi-variable client requests rapidly.

```
None
+-----------------------------------------------------------------------+
|  CONSULTANT INTAKE FORM                                               |
+-----------------------------------------------------------------------+
|  Consultant Name: [ Select / Auto-filled ]                            |
|  Client Name*:    [ Input Text           ]                            |
|                                                                       |
|  Transaction Type: (o) Rent  ( ) Sale                                 |
|  Requirement Type: [ Select: Buying / Renting / Investment / Comm. ]  |
|  Property Type:    [ Toggle: House | Shop | Apartment | Plot | Off. ] |
|                                                                       |
|  Contact Numbers:                                                     |
|    Phone 1*: [ Input     ]  [ + Add Number ]                          |
|                                                                       |
|  Budget Range ($):                                                    |
|    Min: [ e.g. 100,000  ]   Max: [ e.g. 500,000  ]                   |
|                                                                       |
|  Preferred Area / Location:                                           |
|    [ Type & Press Enter (Tags: Downtown x, West End x) ]              |
|                                                                       |
|  Description / Notes:                                                 |
|    [ Multi-line textarea for client specifics...                  ]   |
|                                                                       |
|  Dynamic Custom Fields ("Add Other Options"):                          |
|    Key: [ e.g. Parking Slots ]   Value: [ e.g. 2 ]  [ + Add Field ]   |
|                                                                       |
|  [ ACTION: SAVE & SYNC TO DATABASE                                 ]  |
+-----------------------------------------------------------------------+
```

UI Specifications & Field Logic

1. **Consultant Identity**: Text input or dynamic dropdown synced with `consultants` table. Automatically defaults to currently logged-in user profile.
2. **Client Name**: Standard string input (Required). Validated against empty strings.
3. **Transaction & Property Toggles**: segmented control radio groups for immediate visual feedback (`Rent` vs. `Sale`; `House`, `Shop`, `Apartment`, `Plot`, `Office`, `Other`).

4. **Requirement Type**: Dropdown selecting macro-intent (`Buying`, `Renting`, `Investment`, `Commercial`).
5. **Dynamic Contact Numbers**: Multi-slot input container starting with 1 active field (`Phone 1`). Clicking `+ Add Number` appends `Phone 2` and `Phone 3` dynamically with validation for standardized phone patterns.
6. **Budget / Range**: Dual numerical inputs formatted with currency masks (`Min Budget`, `Max Budget`). Automated logic checks to ensure `Max Budget >= Min Budget`.
7. **Preferred Area / Location**: Tag-style multi-select auto-complete input allowing tokenized entry (e.g., "Downtown", "Marina", "Suburbs").
8. **Description / Notes**: Multi-line auto-expanding text area for contextual unstructured notes.
9. **Dynamic Custom Fields ("Add Other Options")**: Allows agents to add non-standard attributes (e.g., "Furnished: Yes", "Floor Level: 10+") by inserting key-value string pairs stored as JSONB.
10. **Action Trigger**: Primary styled button (`Save & Sync to Database`). Triggers loading spinners, client validation checks, local storage cache backing, dynamic Supabase mutation, and dynamic Toast Notification feedback upon success or failure.

4. Interface 2: Admin Dashboard & Control Center

```
None
+-------------------------------------------------------------------------------
--+
| ADMIN CONTROL CENTER                                  [ Export CSV ] [ Manage
Fields]|
+-------------------------------------------------------------------------------
--+
| FILTERS:
|
| [ Consultant: All ] [ Prop: All ] [ Trans: All ] [ Search Client/Location...
] |
+-------------------------------------------------------------------------------
--+
| CLIENT DATA GRID
|
+--------------+-------------+------------+-------------+--------------+----------
--+
| Client Name  | Type        | Prop Type  | Budget ($)  | Locations    | Actions
|
```

```
+--------------+-------------+------------+-------------+--------------+----------
--+
| John Doe     | Buying      | Apartment  | 250k - 400k | Downtown     |
[Edit][Del]|
| Jane Smith   | Renting     | Office     | 3k - 5k     | Business Bay |
[Edit][Del]|
+--------------+-------------+------------+-------------+--------------+----------
--+
| Inline Editor Active: Click any cell to adjust values & trigger audit log.
|
+-------------------------------------------------------------------------------
--+
```

Key Administrative Functionalities

- **Global Data Grid**: High-density interactive datatable with sticky headers, column sorting, and fixed action columns.
- **Multi-Parametric Filtering**:
    - Filter by Consultant ID
    - Filter by Property Type & Transaction Type
    - Filter by Budget range sliders
    - Text search across Client Name, Notes, and Location tags
- **Inline Dynamic Cell Editing**: Direct mutation of values (e.g., changing lead status from "New" to "Contacted") without modal popups. Changes trigger immediate row lock and persistent database sync.
- **Audit Trail Integration**: Automated shadow logging captures every change event, preserving previous values, updated values, timestamp, and changing user ID.
- **Data Exporter**: One-click extraction module converting the active filtered dataset into formatted CSV/Excel documents.

5. Database Schema Blueprint

```
None
-- Enable UUID extension
CREATE EXTENSION IF NOT EXISTS "uuid-ossp";

-- 1. CONSULTANTS TABLE
CREATE TABLE consultants (
```

```
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    full_name VARCHAR(255) NOT NULL,
    email VARCHAR(255) UNIQUE NOT NULL,
    phone VARCHAR(50),
    role VARCHAR(50) DEFAULT 'staff' CHECK (role IN ('admin', 'staff')),
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 2. CLIENT REQUIREMENTS TABLE
CREATE TABLE clients_requirements (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    consultant_id UUID REFERENCES consultants(id) ON DELETE SET NULL,
    client_name VARCHAR(255) NOT NULL,
    transaction_type VARCHAR(50) NOT NULL CHECK (transaction_type IN ('Rent',
'Sale')),
    requirement_type VARCHAR(50) NOT NULL CHECK (requirement_type IN ('Buying',
'Renting', 'Investment', 'Commercial')),
    property_type VARCHAR(50) NOT NULL CHECK (property_type IN ('House', 'Shop',
'Apartment', 'Plot', 'Office', 'Other')),
    contact_numbers JSONB NOT NULL DEFAULT '[]'::jsonb, -- Stores array of
strings: ["+12345", "+67890"]
    preferred_locations JSONB NOT NULL DEFAULT '[]'::jsonb, -- Stores array of
tags: ["Downtown", "Suburbs"]
    budget_min NUMERIC(12, 2) DEFAULT 0,
    budget_max NUMERIC(12, 2) DEFAULT 0,
    notes TEXT,
    custom_attributes JSONB DEFAULT '{}'::jsonb, -- Stores ad-hoc key-value pairs:
{"parking_slots": "2", "furnished": "true"}
    status VARCHAR(50) DEFAULT 'New' CHECK (status IN ('New', 'Contacted', 'In
Progress', 'Closed', 'Archived')),
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 3. GLOBAL CUSTOM FIELDS DEFINITION TABLE (Optional schema manager)
CREATE TABLE custom_fields_config (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    field_label VARCHAR(100) NOT NULL,
    field_key VARCHAR(100) NOT NULL UNIQUE,
```

```
    data_type VARCHAR(50) DEFAULT 'text' CHECK (data_type IN ('text', 'number',
'boolean', 'select')),
    options JSONB DEFAULT '[]'::jsonb, -- For select dropdown options
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 4. ADMIN AUDIT LOGS TABLE
CREATE TABLE admin_audit_logs (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    requirement_id UUID REFERENCES clients_requirements(id) ON DELETE CASCADE,
    modified_by UUID REFERENCES consultants(id) ON DELETE SET NULL,
    field_changed VARCHAR(100) NOT NULL,
    old_value TEXT,
    new_value TEXT,
    changed_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- INDEXES FOR PERFORMANCE
CREATE INDEX idx_requirements_consultant ON clients_requirements(consultant_id);
CREATE INDEX idx_requirements_status ON clients_requirements(status);
CREATE INDEX idx_requirements_custom_attributes ON clients_requirements USING gin
(custom_attributes);
CREATE INDEX idx_requirements_locations ON clients_requirements USING gin
(preferred_locations);

-- AUTOMATIC UPDATED_AT TRIGGER
CREATE OR REPLACE FUNCTION update_modified_column()
RETURNS TRIGGER AS $$
BEGIN
    NEW.updated_at = NOW();
    RETURN NEW;
    END;
$$ language 'plpgsql';

CREATE TRIGGER update_clients_requirements_modtime
    BEFORE UPDATE ON clients_requirements
    FOR EACH ROW
    EXECUTE PROCEDURE update_modified_column();
```

6. Prompt to Copy-Paste into Claude

n
Create a complete, single-file React component for a Real Estate CRM and Lead Management System using Tailwind CSS and Lucide React icons.

The application must include two main views accessible via a top navigation tab bar:

1. "Consultant Form" (Staff Intake Interface)
2. "Admin Dashboard" (Control Center & Data Grid)

Feature Requirements:

View 1: Consultant Form (Intake)

- **Consultant Selection**: Dropdown to choose active consultant.
- **Client Name**: Required text input.
- **Transaction Type**: Toggle buttons for "Rent" vs "Sale".
- **Requirement Type**: Dropdown (Buying, Renting, Investment, Commercial).
- **Property Type**: Segmented visual selectors for House, Shop, Apartment, Plot, Office, Other.
- **Contact Numbers**: Dynamic input field allowing users to click "+ Add Phone" to add up to 3 separate phone number fields with individual remove buttons.
- **Budget Range**: Dual numerical inputs for Minimum Budget and Maximum Budget with visual currency ($) tags.
- **Preferred Locations**: Input field where pressing Enter adds a location tag/chip (with 'x' to remove).
- **Notes/Description**: Multi-line textarea.
- **Dynamic Custom Fields ("Add Other Options")**: Form section where consultants can enter a "Key" (e.g., "Parking Slots") and "Value" (e.g., "2"), click "Add Field", and display them as active dynamic key-value badges.
- **Submit Button**: "Save & Sync to Database" button that validates required fields, simulates dynamic API submission, displays a temporary inline dynamic success toast, and appends the entry to local state shared with the Admin view.

View 2: Admin Dashboard

- **Stats Summary Cards**: Top row displaying metrics (Total Leads, Active Rentals, Active Sales, Total Volume).
- **Filters Bar**: Search input (by Client Name/Location), Dropdown for Property Type filter, Dropdown for Consultant filter, and Dropdown for Transaction Type filter.

- **Data Table**: Displaying list of submitted leads showing Client Name, Consultant, Property/Transaction Type, Budget Range, Contact Numbers, Locations (tags), Custom Attributes (as badges), Status pill, and Actions.
- **Inline Status Editor**: Clicking a lead's Status pill allows updating it (New, Contacted, In Progress, Closed).
- **Audit Log Modal/Drawer**: Action button per row to view simulated structural audit log history of changes made to that record.
- **Export Action**: Functional button that converts current visible datatable records into a downloadable `.csv` file.

Design & Architecture Specifications:

- Use standard slate/indigo color palettes (`bg-slate-50`, `bg-indigo-600`, etc.).
- Ensure interactive elements feature dynamic focus rings, hover transitions, and clean typography.
- Populate state with 3-4 realistic mock records on initial load so the Admin view is populated immediately.
- Use Lucide React icons (e.g., `Plus`, `Trash2`, `Download`, `Filter`, `CheckCircle`, `Building`, `Phone`, `MapPin`, `User`, `Edit3`).
- Return ONLY the executable code block suitable for immediate preview in Claude Artifacts.